

A KAVERI FREIGHT MYSTERY · CCAR-P DOMAIN 3, TOLD AS A STORY

The Adventure of the Loading Dock

*being a reminiscence of Dr. Watson concerning the week
Sherlock Holmes connected a thinking machine to a real dock,
and the eleven small crimes he uncovered in doing so*



Read it as a story. The lessons are in Holmes's mouth; you will find
you have learned them without noticing, which is the only way that lasts.

CHAPTER 1

The Crane With No Cable

in which Meera Nair brings a problem that is not yet a problem

It was on a Friday evening in late September, the monsoon having at last withdrawn from Pune, that I found Sherlock Holmes standing at the window of our rooms with his hands clasped behind him, watching the traffic on the Hinjawadi road as if it were a chessboard.

“We are about to receive a visitor, Watson,” said he, without turning. “A woman of authority, in a hurry, who has just won an argument and is beginning to suspect she should not have.”

“You can see all that from a window?”

“I can see a company car idling with its engine running, which means the driver expects to leave soon; and a woman on the pavement reading a document for the third time, which means the document has been approved and she is now wondering what she has approved. Come in, Ms. Nair.”

Meera Nair, Head of Operations at Kaveri Freight & Fulfilment, did not waste our time. Her board had that afternoon approved a magnificent thing: an artificial-intelligence control tower that would sit above forty dispatchers and four hundred rulebooks and answer, in seconds, every question a customer could ask about a parcel. The plans were beautiful. The vendor was delighted. The developer, a young man named Arjun Bose, was already at his desk.

“And your difficulty?” said Holmes.

“I have no difficulty,” said she. “That is what troubles me. Everyone tells me it worked perfectly in the demonstration.”

Holmes smiled — the thin smile I had learned to dread on behalf of whoever had made him smile.

“Ms. Nair, on Monday morning your crane will stand on the dock, gleaming, and it will be connected to nothing. No power. No rulebooks. No gate passes for the men who drive it. No cameras to see it work. A

crane that lifts nothing is a sculpture. The week that turns a sculpture into a machine is called *integration*, and it is unglamorous, and it is where real systems die. That is the difficulty you have not had yet.”

“The demonstration —”

“— had one user, ten documents, one key taped over every door, and a developer watching the screen. Your dock has forty dispatchers, four hundred documents, three carriers, and nobody watching. Every trouble you will have this week lives in the distance between those two sentences. I shall walk your dock with Dr. Watson, one station at a time, and I shall show you eleven small crimes before they are committed. Shall we begin on Monday?”

She looked at him for a long moment. “The four most expensive words in logistics,” she said slowly, “are *it worked in the demo*.”

“Ms. Nair,” said Holmes, “you may keep that line. I intend to use it on the examiners.”

CHAPTER 2

The Case of the Cancelled Bill

in which a rulebook is cut with a tape measure, and a customer pays for it

Monday came, and with it the first crime, and it was committed against a customer named Padma Deshmukh.

Her consignment, KF-2481, was stuck in customs because a form called an e-way bill had gone missing. She telephoned at twenty minutes to twelve and asked the dispatcher, a capable woman named Sunita, how to re-file it. Sunita asked the machine. The machine, in a warm and confident voice, read out the steps for *cancelling* an e-way bill. Padma, who trusted it, followed them. By noon she had a cancelled bill, and a one-day hold had become a week.

“Nothing was wrong with the machine,” said Arjun, white-faced, when Holmes arrived. “It said exactly what it found.”

“Then let us look at what it found,” said Holmes, and he asked to see not the answer but the shelves.

I should explain, for the benefit of readers who have not lived through such a week, what shelves have to do with it. A thinking machine cannot read four hundred rulebooks every time a question is asked; it would take an hour and cost a fortune. So the rulebooks are cut into pieces beforehand, and the pieces are filed, and when a question comes the machine is handed only the few pieces that answer it. The cutting is done once, weeks before anyone asks anything, by a developer who is thinking about neatness.

Arjun had cut every rulebook into equal pieces of five hundred words, like a man with a tape measure.

“Watson,” said Holmes, “imagine a warehouse that saws every incoming item into equal half-metre boxes. Beautiful, uniform shelves. And a sofa in six boxes on four aisles. When the order for the sofa comes: three trips, and a missing armrest.”

He laid SOP 214 on the desk and drew a line through it with his finger. “Section three is *cancellation*. Section four is *re-filing*. Both say the words

e-way *bill* on every line. Mr. Bose's tape measure cut the book so that the end of section three and the beginning of section four fell into the same box, and the search — which can only judge a box by the words in it — could not tell one procedure from the other. It handed Sunita the box that began mid-cancellation. She read it out. Padma cancelled her bill.”

“How should it have been cut?” asked Meera.

“Along the seams. Every rulebook has headings; a heading is the author telling you where one meaning ends and another begins. Cut there. One section, one box. And — this is the small trick that carries the whole case — write the section's name *inside* the box, on the first line: *SOP 214, section four, re-filing, version seven*. A box that knows where it came from is found more easily, is trusted more easily, and can never be quoted as the wrong section, because it says its name aloud.”

“And the ones that fall across a seam anyway?”

“Let each box carry the last few lines of the box before it — a photograph of its neighbour. It costs a little duplicate storage, so it is a dial one turns, not a habit one keeps. And re-cut every week, because the rulebooks change every week; a search that quotes a repealed rule with confidence is a liar with a citation.”

Meera wrote for some time. “You are telling me,” she said, “that the quality of every answer this machine will ever give was decided six weeks ago, by how a young man cut paper.”

“I am telling you that the ceiling of a search is set when you shelve, and that nothing downstream can raise it. Everything after can only lose height. It is the first law of this dock, Ms. Nair: *chunk along meaning, not length*. Padma paid for the lesson. We shall not make her pay twice.”

CHAPTER 3

The Book Too Small for a Library

in which Arjun, having been burned, reaches for the wrong extinguisher

A man who has just been shown his error will usually commit its opposite before nightfall, and Arjun was no exception. By Monday evening he was proposing to cut, file and search *everything* — including the carrier rate card and six message templates that every dispatcher used on nearly every call.

Holmes asked him one question. “How large is it?”

“Thirty-eight thousand tokens, sir. It changes once a quarter.”

“Then do not search it. Hand it to the machine whole, every time.”

Arjun protested that this was exactly the wastefulness he had been warned against.

“A librarian,” said Holmes, “does not catalogue a three-page leaflet. She pins it to the wall. She catalogues the four-hundred-volume archive. The difference is size, and how often the thing is needed. Below roughly two hundred thousand tokens — call it five hundred pages — a stable document that every call needs should simply ride along in the prompt, and there is a device called *prompt caching* that makes the machine's repeated reading of it cost about a tenth of the first reading. Your four hundred rulebooks are one million four hundred thousand tokens; they need the library. Your rate card needs a drawing pin.”

“So before building the library —”

“— you ask whether the book is big enough to need one. It is the question before every question in this domain, and the examiners love a candidate who forgets to ask it, because he will build a beautiful search engine for a leaflet, and they will mark him down for over-engineering while he is still admiring his work.”

CHAPTER 4

The Register and the Old Foreman

in which two ways of finding a thing each fail on a different day

Tuesday brought two questions a minute apart, and a mystery that I confess I should not have solved in a year.

The first dispatcher typed the exact words *e-way bill re-filing*. The machine found section four at once. The second typed, from a customer's own mouth, *my parcel is stuck and the customer is furious, what do I do*. The machine found nothing useful at all, because no rulebook in Kaveri's possession contains the words *stuck* or *furious*.

Arjun, who was quick, changed the search engine to a cleverer kind. Now the second question was answered — and the first began to stumble.

"I have seen this before," said Holmes, "in a warehouse with two ways of finding a box. There is the *register*: a ledger of exact words. Ask it for *e-way bill* and it returns every box whose label contains those words, and nothing else. It is precise, literal, and blind to synonyms; ask it for *stuck* and it shrugs. And there is the *old foreman*, who has walked the aisles thirty years and files by meaning. Ask him for *customs paperwork problem* and he points at the right aisle though no label says those words. He is fuzzy; he will sometimes point at the plausible-but-wrong aisle; and he is hopeless with an exact code — to him, HSN 8471 and HSN 8473 are neighbours."

"In the machine," I ventured, "who are the register and the foreman?"

"The register is keyword search — the old, exact kind. The foreman is what they call semantic search: every piece of text is turned into a long list of numbers that places it on a map of meaning, the question is placed on the same map, and the nearest pieces win. Each is excellent. Each fails alone — and here is the cruelty of it, Watson — *silently*, on a particular kind of question, so that a demonstration which only ever asks one kind of question never shows the failure."

"Then which should Arjun use?"

“Both. Ask both; trust where they agree. When the question quotes a code, lean on the register; when it describes a symptom, lean on the foreman. Fetch about a hundred and fifty candidates cheaply, let a slower and smarter reader re-sort them, and hand the machine the best twenty. And do the other thing we learned yesterday: write into every box a sentence saying where it came from, so that both the register and the foreman can see it. The people who make these machines measured it — with that sentence, with both finders, and with the re-sorting, the failures fell by two thirds.”

Arjun did it that afternoon. The two questions were answered within the hour, and I noticed that the answer to the first was section four, whole, with its name on its first line.

CHAPTER 5

You Don't Ask the Foreman for Bay Numbers

in which a number is invented, and nobody notices for a day

On Wednesday a manager asked the machine how many shipments had run late that week, and the machine replied: *about thirty-seven*, and cited two rulebooks. The figure went into the Tuesday report.

The true number was forty-one. Nobody had counted anything.

“This,” said Holmes, when the report reached him, “is the most dangerous failure in the whole domain, and I want you to see why. An error stops a man; he asks a human. A fluent, cited, wrong number stops nobody. It walks into a report and lives there.”

“But how did the machine arrive at thirty-seven?”

“It went to the library. It found paragraphs that *mention* lateness — a rule about late fees, a template for apologies — and, being a machine that composes, it composed a plausible number from them. It did precisely what it was built to do. It was sent to the wrong desk.”

He took a pencil and wrote three questions on Arjun's whiteboard.

“*Where is KF-2481 right now?* That is a bay number. You look it up on the tracking board, where the answer sits exactly; you do not ask the foreman to reminisce about it. Search for things *like* the answer and you have turned a certainty into a guess. *How many shipments ran late?* That is a sum over rows; the accountant does it; a library cannot count. *How do I re-file an e-way bill?* That is the one — the only one of the three — that wants the register and the foreman.”

“Three questions, three machines.”

“And a cheap clerk at the front door who reads the *shape* of the question — an id, a counting word, or a how-do-I — and sends it to the right one. The fashionable answer is to pour everything into the clever search and let it sort itself out. It is the fashionable answer because it sounds modern. It is wrong because retrieval follows the shape of the data, not

the fashion. That is the third law. Write it under the first.”

CHAPTER 6

The Pass That Opened Every Bay

in which every login works perfectly, and that is the crime

It was during the Saturday audit that Holmes found what he afterwards called the ghost.

Every caller to the machine — the dispatchers, the refund desk, the carriers' portals — proved its identity flawlessly. Holmes was not interested in the proofs. He asked to see the *permissions*: what each caller, having proved who it was, was allowed to touch. Arjun produced them, a little puzzled, from three places at once and partly from memory.

Holmes ran his finger down the list and stopped at the carrier portal's entry. It had been issued eighteen months earlier, in the days of the demonstration, “to get things working.” Under permissions it said a single character: an asterisk. Everything.

“At a dock gate,” said Holmes quietly, “the guard checks two papers. The gate pass proves who you are. The manifest says which bays your truck may enter. The heist crews of this world are rarely stopped by the first paper; they walk in with a real pass whose manifest nobody narrowed. This portal has needed to read shipment statuses, and nothing else, for eighteen months. In all that time it could have read every customer's records and issued refunds, and every single time it would have authenticated perfectly.”

“Nothing has happened,” said Arjun.

“The audit *is* what happened. Waiting prices the next heist.”

Meera asked what could be done at once, and here Holmes did something I have seen him do only in his most serious cases: he refused two fixes before offering one.

“You will be told to rotate the key. A fresh key to every bay is still a key to every bay. You will be told to write into the machine's instructions, *never issue a refund from the portal*. An instruction can be talked past; the machine can be persuaded, and a persuaded machine with a wide

manifest does the wrong thing sincerely. Neither of those is a lock. A lock is this: write the table down — every caller, the bays it holds, the one bay each door requires — and check it *in the code, before the tool runs*, keyed on who is asking. One identity per caller, never a shared one, or your ledger will read *somebody* on every line. No asterisks. And log every decision — *especially the refusals*. A cluster of refusals at three in the morning, with everything else normal, is somebody trying the doors.”

The fix, when Arjun made it, was one line. The asterisk became *status:read*. That night at three the portal tried the refund door, was refused, and the refusal was written down. Holmes, reading it on Sunday morning over his coffee, looked more pleased than I had seen him all week.

CHAPTER 7

The Toolbelt That Dragged

in which generosity is priced, and found expensive

Arjun's proposal for the machine's tools ran to nine: shipment status, carriers, read a rulebook, look up an order, book a slot, weather, currency, send a text, escalate to a human. It looked generous. Holmes asked him to price it.

"Price it?"

"Every tool you give the machine is a card — its name, what it does, what it needs. The machine must read every card before it chooses. And the cards ride inside every single request, whether the tool is used or not. Count the tokens."

Arjun counted. Nine cards: two thousand three hundred and fourteen tokens a call. The three tools that last week's logs said were actually used: eight hundred and eleven.

"Fifteen hundred tokens of luggage," said Holmes, "times forty thousand calls a day. Sixty million tokens a day, for tools nobody called. Most people guess that figure three times too low. And that is only the first bill. The second is worse. *Look up an order* and *get shipment status* — two cards that say nearly the same thing. Seven times in a hundred the machine picks the wrong one. A new fitter wears every tool the shop sold him; by noon his belt drags; by Tuesday he reaches for the wrong wrench because six look alike. The old fitter carries three and finishes first."

"Then cut to three," said Arjun.

"Not so fast. *Escalate to a human* was used twice last week — both times on refund disputes above ten thousand rupees, on the two worst days of the week. By the count it belongs in the bin. By consequence it is the most important tool on the belt: the day it is needed pays for the year. Trim by the logs *and* by consequence. Merge the lookalikes or sharpen their cards until no one could confuse them — the card is the only thing the machine reads. Give each kind of work its own short belt. And add a

tool back the day a log line asks for it, not the day someone imagines it might be useful.”

He paused at the door. “Every tool you add is a door, Mr. Bose. It costs tokens, it costs confusion, and someone can walk through it. Know who holds the key. That is the first law, and yesterday's ghost was the same law wearing a different coat.”

CHAPTER 8

One Container, Not Five Forklifts

in which Holmes discourses on shipping, and refuses a standard

Four programs at Kaveri needed the same shipment tools — the assistant, the control tower, the dispatchers' screen, and Arjun's own editor — and Arjun added a new tool about once a month, which meant editing four programs. Meanwhile a memorandum proposed that the nightly job which asked IndiaPost for one thing, from one place, should be “migrated to the standard, for consistency.”

“Before the shipping container,” said Holmes, “every port needed a custom crane for every ship, and every ship a custom hold for every cargo. The container did not make cargo better. It made every *pairing* cheap: any ship, any crane, any truck. There is a container standard for machine tools now — they call it MCP — and it does the same thing: one way for any thinking machine to ask a server what it offers and use it. Your four programs dock with one server; you add a tool once; all four discover it the next time they ask for the menu.”

“And IndiaPost?”

“One ship, one crane, forever. A container system for a rowing boat is machinery for nothing. Call it directly. If a second caller ever appears, wrap it then. The rule is arithmetic, not fashion: count the clients and count the changes. Both high, use the standard. One client and no changes, weld it and go home.”

Meera asked what a server actually offered, and Holmes, who enjoys furniture, gave her the dock. “Three things. *Tools*, which do something — the forklifts. *Resources*, which are documents to read — the manifests on the rack; reading one should never cost a forklift trip. And *prompts*: the laminated procedure cards on the wall, written once, kept on the server, instead of pasted into forty dispatchers' heads and going stale in each.”

“There is a fourth case,” said Arjun. “Delhivery has a booking agent — a machine of their own that negotiates pickup slots.”

“Then it is not a forklift and you must not press its buttons. It has judgement. You brief it — the consignment, your constraints, how much surcharge you will accept — you cap how long the conversation may run, and you treat what it proposes as a proposal, to be checked. One talks to a negotiator. One does not look it up.”

As we left he added, for my notebook, that the standard had moved that July — a transport called SSE was gone, and two old habits called sampling and logging were deprecated — “and a candidate who quotes last year's notes will be marked by this year's examiner.”

CHAPTER 9

The Express Lane and the Inspection Bay

in which Holmes explains that speed and accuracy are bought, never wished for

Two memoranda arrived on Thursday, and Holmes read them with the expression of a man tasting milk that has turned.

The first proposed “the largest available model everywhere, for accuracy,” at a blended cost of nine rupees forty per case against a ceiling of six. The second proposed adding a clever re-sorting step inside the two-second lane, where the slowest one call in twenty already took one point nine seconds.

“Every dock runs two speeds, Watson. The express lane waves routine trucks through in seconds — a glance, a stamp. The inspection bay takes twenty minutes and catches what a glance cannot. Inspect everything and the dock jams. Inspect nothing and it is robbed. The skill is not choosing a speed. It is sending each truck to the right one, and writing down why.”

“And these memoranda?”

“Each spends the one currency the situation had fixed. There are only two currencies here: accuracy on one side, speed and money on the other, and every setting in the machine — which model, how much to fetch, whether to re-sort, how long to wait — buys one with the other. When a question fixes a deadline, you may not spend time. When it fixes a price, you may not spend money. Nineteen tenths of a second plus two tenths is more than two seconds; the second memorandum breaks a promise to buy an improvement, and the improvement belongs on the slower lane where there is room for it. The first spends the price ceiling to buy accuracy on apologies that cost four rupees when wrong.”

“How would you spend it?”

“Where mistakes are expensive. Refund decisions are nine per cent of the calls and cost eighteen hundred rupees when wrong; put the large model there. Delay apologies are fifty-five per cent and cost four rupees;

the small model will do. The blend stays under six. And the carrier whose service hangs for forty seconds — give it three seconds, one retry, and then show the last known time marked *as of ten-forty*. A hung call must cost a known maximum. An honest stale answer beats an endless wait.”

He took the first memorandum and wrote across the bottom, in his angular hand: *the constraint, the dial, the direction, and a number — or it is a wish with a price tag.*

CHAPTER 10

The Average That Lied

in which a dashboard is green, and three hundred dispatchers know better

On Friday a slide reached Meera's desk: *retrieval, average three hundred and ten milliseconds*, comfortably inside the three-second promise. It was green. Sunita, when I mentioned it to her, laughed without pleasure. "Tuesday afternoons," she said, "one call in twenty sits there for three seconds while the carrier wakes up."

Holmes asked Arjun for the day's log, and Arjun brought forty thousand lines, each a sentence — *retrieved chunks for ticket in 310ms* — and no way on earth to tell which lines belonged to which call.

"Three kinds of watching, Watson, and they answer three different questions. The *register* writes down what happened, one line per event — but as data, not prose, so that it can be added up. The *camera* follows one truck from gate to bay to gate: a single number stamped on every line that truck touches, so its journey can be pulled back out of the pile of forty thousand. The *gauges* on the wall show trucks per hour and waiting time, over time. Lose the register and you cannot say what happened. Lose the camera and you cannot say where the time went. Lose the gauges and you cannot say whether today is normal. At forty thousand a day, unanswerable means unfixable."

"And the green slide?"

"Averages are where the pain of the unlucky goes to hide. Nineteen quick calls and one slow one average to something that looks healthy; the twentieth caller's afternoon is invisible in it. A promise about speed is a promise about the slowest one in twenty — the ninety-fifth percentile — and Kaveri's is two thousand nine hundred milliseconds. The mean lied. It always does."

He had two further warnings, and I set them down because the examiners are fond of both. "Do not put a consignment number on a gauge: that is a new gauge per truck, forty thousand a day, a wall nobody can read. Gauges are for categories; numbers go in the register. And do not let the alarm bell ring thirty times a day on trifles; a crew that

has learned to ignore the bell will sleep through the fire. Ring it for the few things a customer feels, and ring it seldom.”

“There is a proposal,” said Arjun, “to add a dashboard.”

“A dashboard shows what exists. It creates nothing. Stamp the trace number at the door first; write the lines as data; report the slowest one in twenty. Then buy the dashboard. You cannot fix what you cannot see — the fourth law — and a screen full of pretty panels over an empty register is a man admiring his spectacles in the dark.”

CHAPTER 11

The Truck Loaded With the Whole Warehouse

in which Holmes hands the driver a catalogue, and then takes it back

Arjun's first version, in the days of the demonstration, had solved every question by including everything: every tool card, all four hundred rulebooks, in every request. Three hundred and sixty-two thousand tokens, to answer a question that needed one tool and one section. Holmes, shown the figure, did not comment for some time.

"A driver arrives for one pallet," he said at last. "Warehouse A loads his truck with everything he might conceivably need. He drives away slow, overweight, and mostly wrong. Warehouse B hands him a one-page catalogue at the gate; he asks for bay twelve; he gets bay twelve. The catalogue is the modern habit: a light list of tools up front, the full card fetched only when the machine chooses one; the twenty best pieces of the rulebooks, not the four hundred books. Four thousand tokens instead of three hundred and sixty thousand — with one cost the catalogue can never refund: the trip back to the gate. One extra round-trip."

"Then the catalogue always wins," said Arjun, who had by now learned to distrust his own conclusions, and said it as a question.

"No. And this is the exception the examiners test, because everyone learns the rule and forgets the fine print. For a tiny warehouse with three shelves that every driver uses every time, the catalogue is pure overhead. Load the three shelves. Let caching make the repeat readings cost a tenth. A small, stable, always-used thing rides along whole; a large, changing, sometimes-used thing is catalogued and fetched. Send the map first and the territory on demand — the fifth law — *unless the territory is smaller than the map.*"

He added, because Arjun's long investigations had begun to choke on their own history, that the machine's makers now let the server sweep the desk itself on long runs — clearing the oldest tool results when the pile grew past a hundred thousand tokens, keeping the last three, protecting any tool you named, and letting the machine copy essentials

into a notebook first. “It is not free. Sweeping the desk invalidates what was cached from that point; you pay one re-write and then the turns get cheaper. Every device in this domain has a bill, Watson. Naming the bill is what the examiners mean when they write *evaluate*.”

CHAPTER 12

Tuesday, Ten Forty-One

in which one ticket goes right, and Watson understands the building

A week and a day after Padma's bill was cancelled, I stood behind Sunita's chair and watched her type: *Where is KF-2481 right now, and what do I do to get it out of customs hold?*

I timed it. Two and six-tenths seconds. The answer cited SOP 214, section four, version seven, by name; gave the current status; and ended, as the laminated card on the server required, without promising a date. Padma's bill was re-filed by five past eleven.

Holmes, at my elbow, narrated the journey as if reading it off the truck's side. The door stamped a trace number. The clerk at the front split the question in two — an id for the tracking board, a procedure for the library. The gate checked *status:read* and wrote its line. The connector fetched the status with a forklift and read the rulebook off the rack without one. The register and the foreman found section four, whole, labelled from inside, re-sorted into the best twenty. The window carried a cached preamble, a four-tool belt, twenty pieces and the question. The middle-sized model drafted; the carrier's clock was capped at three seconds. Nine lines in the register. Forty-seven paise.

“Now take one hop away,” he said, “and ask two questions: what breaks, and when would you find out? Drop the trace number: nothing errors, and you learn of it on the bad day. Skip the gate: the same call succeeds today, and the ghost reads every customer's file at three in the morning. Cut with the tape measure: Padma cancels her bill again.”

“None of them stands alone,” I said, and was surprised to hear it in my own voice.

“None. The cameras are what make the belt audit possible. The gate writes to the register. The shelves decide what the dials can buy. The catalogue works because the connector lists before it calls. That is why the examiners put two objectives in one question, and why the right answer usually honours the hop *before* the one they name. Know the building, Watson. Anyone can learn the rooms.”

CHAPTER 13

What Watson Wrote Down

being the five laws, in the words the crew used

I keep, as my readers know, a notebook, and on the train back from Pune I set down the week in the fewest words I could manage. Holmes read the page, made one correction, and handed it back without comment, which from him is praise.

One. Every tool you add is a door; know who holds the key. A tool costs tokens, costs confusion, and can be walked through. Trim by the logs and by consequence.

Two. Chunk along meaning, not length. Cut on the seams, write the name inside the box, size the box to the question. The ceiling is set at shelving time.

Three. Retrieval follows the shape of the data, not the fashion. An id wants the tracking board; a count wants the accountant; a procedure wants the register and the foreman together. A library cannot count.

Four. You cannot fix what you cannot see. The register says what, the camera says where, the gauges say whether today is normal. Promises are about the slowest one in twenty. Write down the refusals.

Five. Send the map first, the territory on demand — unless the territory is small, stable, always used and cacheable, when you load the truck and let the cache pay.

And beneath them, in Holmes's hand, the correction — which was not a correction but a sixth line: *“It worked in the demo” are the four most expensive words in logistics. Ask what breaks at forty thousand a day.*



“You have learned it, Watson,” said Holmes, as the lights of Lonavala went by. “I watched you. You learned it by being annoyed at Arjun, and then by being sorry for Padma, and then by being pleased for Sunita. That is the only way it stays.”

“And the examination?”

“Will describe the same week, in the dry voice of a man who was not there. Read each question twice — once for what the data *is*, once for what the question *asks* — underline the thing they have fixed, and choose the answer that Meera would sign. The rest is the dock.”